



Air University
Final Semester Examinations: Fall 2025

Student ID: _____

Subjective Part
(To be solved on Answer Books only)

Subject: Computer Networks
Class: BSCYS-FA24
Section(s): A
Course Code: CS260

Time Allowed: 180 Minutes
Max Marks: 100
FM's Name: Ms. Laraib Javed
FM's Signature: *[Signature]*

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.
- Calculators are allowed.

16/12/25

Q. No. 1 (CLO 2)		Marks
A host sends packets to a destination but receives "Destination Unreachable" errors. (a) Identify which layer generates this message and why. (b) Explain how routing failures cause this condition. (c) Analyze how the Data Link Layer still successfully delivers frames locally. (d) Explain how the Transport Layer reacts to such errors.		5+5+5+5
Q. No. 2 (CLO 2)		
A sender uses rdt3.0 (stop-and-wait) over a channel with a round-trip time (RTT) of 40 ms, packet size 8,000 bits, and link rate 1 Gbps. (a) Calculate the sender utilization. (b) Explain why utilization is low despite a high-speed link. (c) How does pipelining (GBN and SR) improve utilization? (d) If the sender switches to a pipelined protocol with a window size of 50, qualitatively explain whether the utilization will approach 100% or not, and identify the remaining factors that may still limit utilization.		5+5+5+5
Q. No. 3 (CLO 3)		
A user reports that a large file transfer between two hosts is extremely slow and occasionally fails. The path includes: • A wireless LAN at the sender side • Multiple routers across the Internet • A wired Ethernet LAN at the receiver side (a) Analyze possible causes of performance degradation at the Physical Layer.		5+5+5+5

- (b) Explain how CSMA/CA at the Data Link Layer may introduce additional delay.
- (c) Analyze how IP best-effort delivery at the Network Layer affects reliability.
- (d) Explain how TCP mechanisms attempt to recover from these issues.

Q. No. 4 (CLO 2)

An organization is assigned the IP block 181.56.0.0/16 (Class B) and requires 1000 subnets.

5+5+5+5

- (a) Design an appropriate subnet mask.
- (b) Calculate the number of usable hosts per subnet.
- (c) Identify the first and last usable IP addresses in the third subnet.
- (d) Explain the trade-off between the number of subnets and the number of hosts per subnet in this design, and discuss how this trade-off impacts network scalability

Q. No. 5 (CLO 3)

Scenario: Security Assessment for TechCorp's Web Application TechCorp, a data management company, has hired your penetration testing team to evaluate the security of their newly developed web application. Your primary focus is the login mechanism and database security.

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Consider a scenario where a company uses email services to communicate both internally and externally. Design a diagram showing the interaction between different email components, such as the Mail User Agent (MUA), Mail Transfer Agent (MTA), and Mail Delivery Agent (MDA). Explain how the protocols SMTP, POP3, and IMAP are used in this communication flow. Discuss the challenges faced when handling large volumes of emails, such as spam filtering, and describe how different agents (MTA, MDA) handle these challenges.